Programming Aspects of Different types of Matrices

Matrix

- Row vector,
- Column vector,
- Full Matrix
- Symmetric matrix,
- Sparse matrix,
- Symmetric sparse matrix.

Full (Square) Matrix ===>vector

- $A = \begin{matrix} 12 & 62 & 63 \\ 43 & 58 & 67 \\ 17 & 84 & 69 \end{matrix}$
- Row vector ====>
- B = [12, 62, 63, 43, 58, 67, 17, 84, 69]
- Column Vector==>
- C = [12, 43, 17, 62, 58, 84, 63, 67, 69]
- Row vector
  - A[i,j] = B[i*n +j] (Java)
- Column Vector
  - A[i,j] = C[(j)*n+i] (Java)

  It should be noted that in java the indices of arrays ranges from 0 to (n-1) instead of (1-n)

Assignment 03

1. Write a Java program to convert a general matrix A (m*n) into row vector and column vector?

   - Read the number of rows
   - Read the number of columns
   - Read and store the elements
   - Display in the matrix form
   - Display in the Row Vector form
   - Display in the Column Vector form
   - Find the location of the row vector where the element of the full matrix A(i,j) is stored. Where i – row number, j – column number
   - Find the location of the column vector where the element of the full matrix A(i,j) is stored. Where i – row number, j – Column Number

- $A = \begin{matrix} 1 & 5 & 9 \\ 3 & 5 & 4 \\ 7 & 8 & 2 \end{matrix}$ B = [1, 5, 9, 3, 5, 4, 7, 8, 2]

- A – 3x3; A[1,2] = 4
- A[1,2] = B[k]; k=?
  - A[i,j] = B[(i)*n-1 +j+1] = B[i*n +j]

  i=1; j=2

  K= i*n+j = 1*3+2 = 5

$$A = \begin{bmatrix} 3 & 4 & 5 \\ 6 & 5 & 4 \\ 9 & 7 & 3 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 3 & 6 & 9 \\ 4 & 5 & 7 \\ 5 & 4 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 4 & 5 \\ 4 & 15 & 7 \\ 5 & 7 & 3 \end{bmatrix}$$

Symmetric Matrix

- Row ==> i
- Col ==> j
- (i-1)(i)/2 + i + j

- Check with
- (i,j) == > (2,1) (3,1), (3,2)

Matrix Abstract

```
public abstract class Matrix
{
        protected int columns;
        protected int rows;

        public int rowSize()
           { return rows;}

        public int columnSize()
           { return columns; }

        public abstract void display();
        public abstract double elementAt(int i, int j);
        public abstract Matrix multiply(double K);
        public abstract Matrix multiply(Matrix B);
        public abstract columnVector multiply(columnVector B);
        public abstract Matrix add(Matrix B);
        public abstract Matrix sub(Matrix B);
        public abstract void swapRows(int i, int j);
        public abstract void swapColumns(int i, int j);
        public abstract Matrix transpose();
        public abstract Matrix inverse();
}
```

Exercise

- Complete the Java code for the class matrix shown in the previous slide.

- General Matrix m*n
  - Number of rows: m
  - Number of Columns: n
- Row vector
- Column vector
- Symmetric matrix with row vector

Symmetric Matrix

- $A = \begin{matrix} a & b & c \\ b & d & e \\ c & e & f \end{matrix}$
- If A[i,j] = A[j,i] for all i and j then it is called Symmetric Matrix

Symmetric Matrix

- How to represent?
  - Rowvector
  - Number of elements
- Method “ElementAt()” – write??
- Constructors
  - symmetricMatrix(int Matsize)
  - symmetricMatrix(rowVector A, int Matsize)
  - symmetricMatrix(double[] A, int Matsize)
  - symmetricMatrix(String datafile, int Matsize)
  - symmetricMatrix(Matrix A)

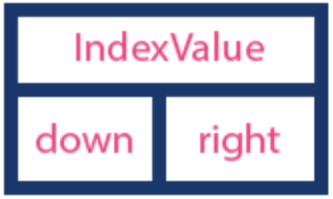
Sparse Matrix

What is meant by sparse matrix?

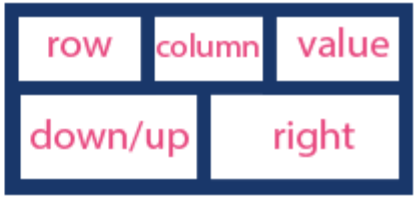
a sparse matrix is a matrix in which most of the elements are zero.

Representing Sparse Matrix

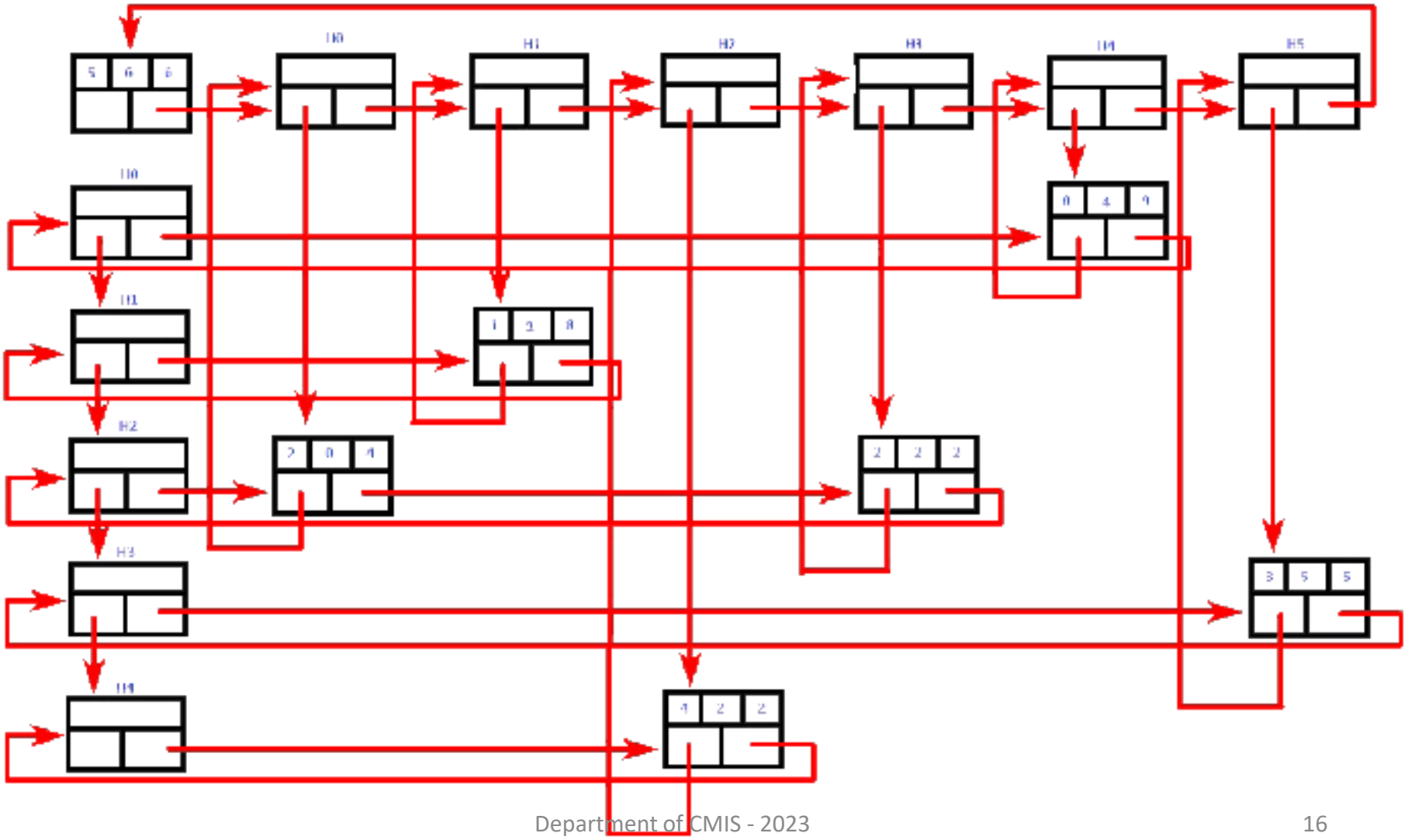
Header Node



Element Node



Representing Sparse Matrix



Sparse Symmetric Matrix

- rowVector a; // elements
- integerVector diagonal; //location of Diagonal elements
- integerVector columnNums; // column number of each element
- int nElements; // Number of elements

- How to write method
  - elementAt(int i, int j)